

Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Charishmakongara/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	Data preprocessing, Crop prediction using Logistic Regression, K-Means clustering analysis, User input form, Real-time crop recommendation, Model saving using Pickle
Pending / Incomplete Features	Advanced weather API integration, Fertilizer recommendation, Cloud deployment (if not implemented)
Setup / Run Instructions	1. Install Python and required libraries. 2. Install dependencies using <code>pip install -r requirements.txt</code>. 3. Run the Flask application using <code>python app.py</code>. 4. Open the application in a web browser using the local Flask URL. 5. Enter soil and environmental parameters to receive the recommended crop.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop: Smart Agricultural Production Optimization Engine was successfully developed using GoogleColab for Machine Learning model development and Flask for web application deployment. The system accurately predicts the most suitable crop based on user-provided soil nutrient and environmental parameters. The modular code structure improves readability, simplifies maintenance, and supports future enhancements such as weather API integration, fertilizer recommendations, and cloud deployment.